A Frozen, Falsifiable Prediction for the Fine–Structure Constant from Photon–Recoil Measurements

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Abstract

This note states a single, quantitative prediction: the values that a photon–recoil measurement of the fine–structure constant α is allowed to return. The prediction is dated ahead of the data that will test it, it is sharp enough to be wrong, and it is *self–contained*: everything needed to test it is on this page, and you do not need to accept the theory that produced it. The claim, the numbers, what already fits, what is genuinely blind, and what would kill it are all stated below.

1 The prediction, in one line

A photon–recoil determination of α^{-1} is predicted to return not an arbitrary number, but one of a small *discrete* set: a fixed reference value plus a step of fixed size times one of three allowed weights,

$$\boxed{\alpha^{-1} = \alpha_{\rm ref}^{-1} + w\, f_1, \qquad w \in \{-0.7236,\ 0,\ +0.2764\}} \tag{1}$$

with the reference value and the step size

$$\alpha_{\rm ref}^{-1} = 137.0359991703, \qquad f_1 = \varphi \times 10^{-7} = 1.6180 \times 10^{-7}, \tag{2}$$

where $\varphi = (1+\sqrt{5})/2 = 1.6180\ldots$ is the golden ratio. Numerically, the only allowed outcomes are the three values

weight w	offset $w\, f_1$	predicted α^{-1}
-0.7236	-1.171×10^{-7}	137.035999053
0	0	137.0359991703
$+0.2764$	$+4.47 \times 10^{-8}$	137.035999215

The blind, falsifiable content is the discreteness. A species that has not yet been measured must land on one of these three values, not somewhere in the continuum between them. Nothing about any measured value of α was used to build this lattice of allowed values.

2 Where the three numbers come from

Only the ingredients are given here, not the derivation — the prediction is tested on its numbers, not on the theory behind them. The step size is $f_1 = \varphi \times 10^{-7}$. The two non–zero weights are the two closed forms

$$-\frac{\varphi}{\sqrt{5}} = -0.7236, \qquad 1 - \frac{\varphi}{\sqrt{5}} = \frac{1}{\sqrt{5}\,\varphi} = +0.2764, \tag{3}$$

and the third allowed weight is exactly zero. These three fix the lattice with no free parameter and with no atomic input. Two consequences are worth noting, since they are extra predictions on data yet to come: the two non–zero values are separated by exactly one step, $|-0.7236 - (+0.2764)| \cdot f_1 = f_1$, and their sizes are in the ratio $0.7236/0.2764 = \varphi^2$.

3 What is already known: it survives, but this is not yet a confirmation

The two best existing photon–recoil determinations are

	current best value	predicted value
^{87}Rb (Morel *et al.*, *Nature* 2020)	137.035999206(11)	137.035999215
^{133}Cs (Parker *et al.*, *Science* 2018)	137.035999046(27)	137.035999053

Assigning rubidium to the $+0.2764$ value and caesium to the -0.7236 value, both survive: rubidium sits 0.82σ and caesium 0.27σ from their predicted values.

The honest caveat, stated plainly. *Which* species was assigned to *which* of the two weights was chosen using these same 2018/2020 numbers. So the good agreement of the two *known* points is partly built in — it is not an independent check on past data. Two things, and only these two, are genuinely blind and untested: (i) the *discreteness* for any new species, and (ii) the *direction* the two known values will move as their precision improves. This note is a dated, conditional forecast, not a derived prediction; the single missing step that would make it fully blind is a rule that fixes species–to–weight *before* the data, which is not claimed here.

4 The decisive test, with the number to beat

The prediction is signed, and the signs are the exposure: the true rubidium value is predicted to sit *above* its current central value, by $+9 \times 10^{-9}$ in α^{-1}, and the true caesium value likewise above its current central value, by $+7 \times 10^{-9}$. The test becomes decisive at three standard deviations once a recoil measurement reaches an uncertainty near 3×10^{-9} on α^{-1} — a relative precision of 2×10^{-11}, about four times better than the 2020 rubidium result, on the trajectory that programme is already on. No new kind of apparatus is required; the test is a precision improvement on a measurement that already exists.

5 What kills it

1. **Convergence.** If improved recoil determinations move together toward a single common value, rather than sorting onto discrete steps, the prediction is dead.
2. **Off–lattice.** If any new species lands off the set $\{-0.7236, 0, +0.2764\}$ (in units of f_1 from the reference) at three standard deviations, the prediction is dead.
3. **Wrong sign.** If the improved rubidium value drifts *away* from 137.035999215 at three standard deviations, the assignment fails and the prediction is dead.

6 What this is not

This is *not* a claim that α varies in space or time: the reference value does not drift, and only the apparatus–dependent step changes, so existing atomic–clock limits on a temporal drift of α are consistent with it and do not test it. The competing ordinary explanation is left open and not treated as excluded: an unrecognised systematic in one interferometer could produce a rubidium–caesium offset with none of the structure predicted here. A ^{229}Th nuclear clock is deliberately *excluded* from this lattice, because it measures a different quantity (a frequency sensitivity, not an absolute h/M recoil comparison) and does not belong on the same axis. Finally, if this prediction fails, what falls is bounded: it kills this recoil claim, and it does not by itself confirm or refute the broader body of work it was extracted from.

Freeze and provenance

The values were constructed on 4 July 2026, ahead of any photon–recoil or nuclear–clock determination of α published after that date, and are frozen. They are extracted from the author's larger work (Quantum Gauge Theory, Second Edition, revision r26.9), but the prediction above stands on its own: none of that theory needs to hold for the test to be run. After deposit in a time–stamped, immutable record, the frozen content may change only to register a PASS, FAIL, or INCONCLUSIVE verdict of the data — never its numerical content.

Machine–checkable record of the frozen numbers: `qgt_branch_prediction_frozen.py`, SHA-256 `fb68dbbb869f5ade...56ce36c1`, which recomputes every value above from the underlying formulas rather than hard–coding them. *Data used for the current–value comparison:* Morel *et al.*, *Nature* **588**, 61 (2020) [Rb]; Parker *et al.*, *Science* **360**, 191 (2018) [Cs].